

# Higher-Order Epistemic Networks for Navigating the Archive of Formal Proofs

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**Abstract.** Formal proof libraries such as the Archive of Formal Proofs (AFP) for Isabelle/HOL form directed networks of machine-verified deduction. Existing neural provers rely on flat retrieval over unpartitioned text or pairwise citations, causing search distraction and context exhaustion. Drawing on higher-order networks, we model formal libraries as Embedding-driven Epistemic Landscapes (EEL), representing theorems as 4-aspect 3-simplices spanning premises ( $P$ ), proof methods ( $M$ ), formal steps ( $F$ ), and implications ( $I$ ). Navigation proceeds via twelve directional transition operators between aspects, prioritised by Landscape Height centrality over the citation DAG. We implement this architecture as an open tool service (I/L, Isabelle/Landscape) using the Model Context Protocol (MCP). Across 140 formal theorems, simplicial navigation solves twice as many discordant theorems (8 vs. 4) as monolithic retrieval, reducing token expenditure by 16.5%. These results demonstrate that higher-order representations suppress cognitive search friction while improving formal proof synthesis.

**Keywords:** Higher-Order Networks, Epistemic Landscapes, Theorem Proving, Isabelle/HOL, Archive of Formal Proofs.

## 1 Introduction

Formal mathematical repositories such as the Archive of Formal Proofs (AFP) for Isabelle/HOL [12, 22], the Lean Mathematical Library (*Mathlib*), and the Coq library constitute massive, machine-checked corpora of human and algorithmic deduction. In interactive theorem provers (ITPs) like Isabelle/HOL, mathematical theories and program verifications are checked against a trusted logical kernel, compiling deductions into a verified archive. Unlike empirical citation networks shaped by social prestige, visibility dynamics, and noisy attribution [14, 9], formal mathematical corpora define strictly sound, deductive directed acyclic graphs (DAGs). In these graphs, an edge between two lemmas represents a machine-checked logical dependency. Viewed through the lens of network science, formal libraries are exemplary realisations of *complex knowledge networks*, exhibiting multi-scale hierarchy, clustering, and heavy-tailed connectivity.

Concurrently, scientific and mathematical problem-solving has long been conceptualised as navigation across an *epistemic landscape* [21], where agents explore a topology of research *approaches* characterised by multiple epistemic *aspects* (such as problems, methods, and outcomes). Recent advances show that dense neural embeddings can reconstruct the continuous topography of epistemic landscapes from unstructured scientific discourse [6]. When applying autonomous AI agents and Large Language Models (LLMs) to formal proof synthesis [17, 8], however, conventional retrieval mechanisms encounter critical limitations.

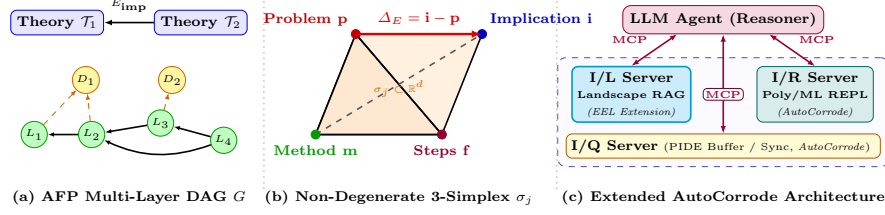
**The Limits of Pairwise Backbones and Flat Retrieval.** In complex networks science, Yassin et al. [24] recently demonstrated that conventional pairwise similarity backbones filter networks by evaluating purely dyadic relationships, often destroying multi-node topological organisation and reachability. In contrast, higher-order backbones derived from simplicial complexes preserve multi-node topological reachability and retain structurally unique simplices [24, 1, 3].

In formal deduction, a mathematical theorem cannot be reduced to a single point or a simple dyadic citation  $u \rightarrow v$ . A citation edge merely indicates that  $u$  depends on  $v$ , but collapses *which aspect* of  $v$  is invoked: its premise hypotheses, its overarching induction strategy, its intermediate calculation steps, or its terminal conclusion. Treating formal libraries as flat pairwise graphs or unpartitioned text strings discards this internal polyadic geometry. Recent automated theorem proving systems mirror the flaws of pairwise backbones by treating formal mathematics either as flat natural-language text or as disconnected dyadic dependencies:

**IsaSearch** [11] constructs an embedding-based search engine over the AFP by generating natural-language informalisations of theorems. However, *IsaSearch* is designed solely for passive human search, remains completely blind to the underlying citation DAG, and conflates distinct epistemic aspects (premises, intermediate assertions, tactics, and automation directives) into monolithic document strings.

**IsabeLLM** [10] implements automated theorem proving in Isabelle by combining LLM generation with Retrieval-Augmented Generation (RAG). *IsabeLLM* retrieves top- $k$  nearest lemmas from vector databases and injects their raw formal text into the prompt. Crucially, Jones and Knottenbelt [10] explicitly report that flat monolithic RAG breaks down on multi-theory verifications because injecting uncured proof text rapidly exhausts the model’s context window. They conclude that dynamic tool-use and modular interfaces (such as the Model Context Protocol) represent an urgent, unfulfilled requirement for neural theorem proving.

**The Isabelle Assistant and the I/L Ecosystem.** To enable AI agents to interact with formal proof environments dynamically without prompt exhaustion, the Model Context Protocol (MCP) provides an open standard for modular tool invocation. The open-source **Isabelle Assistant** (part of AutoCorrode [2]) provides two foundational servers: the **I/R** (Isabelle/REPL) server for



**Fig. 1.** Architecture of the I/L ecosystem. (a) Multi-layer formal DAG  $G$  spanning theories  $\mathcal{T}$ , definitions  $\mathcal{D}$ , and lemmas  $\mathcal{L}$ ; (b) Non-degenerate epistemic 3-simplex  $\sigma_j$  in  $\mathbb{R}^d$  with net epistemic expansion  $\Delta_E$ ; (c) Extended AutoCorrode agent orchestration over the Model Context Protocol.

rapid speculative tactic execution in a live kernel, and the **I/Q** (Isabelle/Query) server for editor buffer inspection. Our central contribution is the **I/L** (Isabelle/Landscape) server: a new, third MCP server that extends this ecosystem with higher-order simplicial retrieval over the EEL complex of the AFP, allowing the agent to query strategic and tactical aspects independently.

Simplicial complexes have previously been applied in the science of science to study scientific collaboration and concept co-occurrence [16, 19]. In those studies, however, vertices represent authors or keywords, and simplices represent document co-authorship cliques or shared vocabularies. Our construction is fundamentally different in kind: **vertices represent distinct epistemic discourse roles ( $P, M, F, I$ ) within a single formal artefact**, capturing the internal deductive geometry of proofs. The global simplicial complex is then constructed by assembling these intra-artefact simplices via conditional transition distribution operators and logical citation DAGs.

**Contributions.** In this paper, we develop a higher-order complex network framework for formal mathematical repositories and make three primary contributions:

1. **EEL Instantiation for Isabelle/AFP:** Building on the EEL framework [7, 6], which models scientific discourse through generic epistemic aspects ( $P, M, F, I$ ), we provide an **Isabelle/HOL-native reinterpretation** suited to formal deduction: premises ( $P$ ), proof methods ( $M$ ), tactical formal steps ( $F$ ), and implications ( $I$ ). We interpret directional cross-aspect transition operators  $D(Y|x)$  as **Generalised Gluing Operators**, assembling individual 3-simplices into a navigable higher-order epistemic network.
2. **Topological Centrality via Landscape Height ( $H$ ):** We define Landscape Height  $H(v)$  on the transpose citation graph  $G^T$  to quantify the downstream deductive reachability cone of each theorem, establishing a fused ranking function that suppresses ephemeral leaf lemmas in favor of foundational hubs.
3. **I/L: Extending Isabelle Assistant with Simplicial RAG & Live Evaluation:** We implement **I/L** (Isabelle/Landscape) as a modular MCP server extending the Isabelle Assistant ecosystem (AutoCorrode [2]). In a live benchmark of 140 formal theorems across 420 trials, simplicial navigation

solves twice as many discordant theorems as flat monolithic RAG, while reducing completion token expenditure by 16.5% ( $p = 0.00014$ ) through search distraction suppression.

## 2 Embedding-Driven Epistemic Landscapes for the AFP

We formalise the AFP as a directed multi-layer graph  $G = (V, E)$ , where the node set  $V = \mathcal{T} \cup \mathcal{D} \cup \mathcal{L}$  partitions into theories  $\mathcal{T}$ , definitions  $\mathcal{D}$ , and theorems/lemmas  $\mathcal{L}$ . The edge set  $E = E_{\text{import}} \cup E_{\text{cite}} \cup E_{\text{def}}$  encompasses three relational layers: (i) **Theory Imports** ( $E_{\text{import}} \subseteq \mathcal{T} \times \mathcal{T}$ ), the macroscopic DAG of modular encapsulation; (ii) **Lemma Citations** ( $E_{\text{cite}} \subseteq \mathcal{L} \times \mathcal{L}$ ), where  $(u, v)$  indicates that the proof of  $u$  cites  $v$  via fact chaining (**using, from**) or automation (**by (simp add: v), by (metis v)**); and (iii) **Definitional Dependencies** ( $E_{\text{def}} \subseteq (\mathcal{L} \cup \mathcal{D}) \times \mathcal{D}$ ), grounding theorems and compound definitions to foundational constant declarations. Empirical examination of  $G_{\text{cite}}$  reveals a heavy-tailed in-degree distribution [14]: more than 65% of lemmas are single-use helper lemmas ( $k_{\text{in}} \leq 1$ ), while foundational lemmas serve as high-degree topological hubs.

### 2.1 Epistemic Simplices and Aspect Segmentation

Following Weisberg & Muldoon [21], each coordinate in an epistemic landscape represents a distinct *approach* (in our context, a proving approach identified by the formal theorem being proved), described by multiple *aspects* that jointly define its coordinates. In the EEL framework [7, 6], any formal artefact is modelled as a four-aspect configuration  $(P, M, F, I)$  corresponding to problem, method, finding, and interpretation: a minimal closed inferential cycle grounded in Argumentative Zoning [20] and Peirce’s inferential triad. We retain this arity-4 structure and provide an **Isabelle/HOL-specific reinterpretation** of each role suited to formal deductive mathematics, where aspects are syntactically grounded rather than extracted from prose abstracts:

1. **Premises ( $P$ ):** Antecedent assumptions, sort bounds (`::linorder`), type constraints, and locale contexts.
2. **Proof Method ( $M$ ):** The strategic deductive roadmap (e.g., structural induction, case split, or equational rewriting pipeline).
3. **Formal Steps ( $F$ ):** The tactical machinery discharging intermediate assertions, captured as coupled step maps representing the *execution record* of the proof.
4. **Implications ( $I$ ):** The terminal consequent with its global automation directives (`[simp]`, `[intro]`, `[elim]`).

This arity-4 decomposition fixes the topological dimension of each theorem to an **epistemic 3-simplex** (a tetrahedron with 4 vertices and 6 edges), inducing  $\binom{4}{2} \times 2 = 12$  directional transition operators between ordered pairs of aspects. Each aspect is embedded via  $\mathcal{E} : \mathfrak{T} \rightarrow \mathbb{R}^d$  with  $d = 1024$  (**voyage-code-3**), where  $\mathfrak{T}$  is the space of formal code expressions. The four aspects are *not* four disjoint vector spaces, but empirical submanifolds within a **single shared ambient**

**space**  $\mathbb{R}^d$ , which guarantees that displacement vectors ( $\Delta_{PM} = \mathbf{m} - \mathbf{p}$ ) and cross-aspect operator distributions are mathematically well-defined. The resulting tetrahedron is defined as the **discourse simplex**  $\sigma_j = \text{conv}(\mathbf{p}_j, \mathbf{m}_j, \mathbf{f}_j, \mathbf{i}_j)$ , a geometric polytope in  $\mathbb{R}^d$  encoding multi-aspect relationships.

Dense embeddings of formal code exhibit severe anisotropy: frequent outer syntax boilerplate (`lemma`, `fixes`, `assumes`, `shows`, `:`) dominates the top principal components, collapsing non-linear projections onto a degenerate one-dimensional curve ( $\psi_2 \approx f(\psi_1)$ ). To recover the intrinsic mathematical geometry, we apply *All-but-the-Top* spectral filtering [13]:

$$\mathbf{x}' = \mathbf{x} - \sum_{i=1}^k (\mathbf{x} \cdot \mathbf{u}_i) \mathbf{u}_i \quad (1)$$

where  $\mathbf{u}_i$  are eigenvectors of the corpus covariance matrix  $\Sigma = \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j \mathbf{x}_j^T$ . To establish the threshold, we swept  $k \in [0, 100]$  over 1,924 benchmark theorems: the domain cluster silhouette score surges from  $S_{\text{domain}} = 0.191$  at  $k = 0$  to  $+0.521$  at  $k = 5$ , remaining robust at  $+0.439$  at  $k = 10$  (41.0% variance removed). Beyond  $k = 10$ , mathematical semantics are progressively stripped alongside syntax (domain separation collapses for  $k \geq 15$ ; isotropic noise at  $k \geq 40$ ). We therefore adopt  $k = 10$  as the robust operating baseline (full sweep data are in the Supplementary Material).

The role of I/L’s aspect segmentation engine is to parse each proving approach in the AFP into its four constituent aspect coordinates ( $P, M, F, I$ ) for embedding and indexing. To illustrate, consider a representative lemma from `AVL-Trees.AVL` [15]:

```
lemma avl_insert [simp, intro]:
  fixes x :: 'a :: linorder
  shows "avl t ==> avl (insert x t)"
  by (induction t rule: avl.induct)
    (auto simp: avl.simps height_insert)
```

Our parser extracts the four coordinates as follows:

1. **Premises ( $P$ ):** Type constraints and hypotheses:  
`fixes x :: 'a :: linorder; assumes avl t.`
2. **Strategy ( $M$ ):** The strategic roadmap inferred from the proof:  
`structural-induction on (t) via avl.induct; auto.`
3. **Tactics ( $F$ ):** Coupled step map:  $\text{Step}_1 = \langle \text{avl (insert x t)}, \text{auto simp: avl.simps, \{avl.simps, height\_insert\}} \rangle$ .
4. **Conclusions ( $I$ ):** Consequent with automation directives:  
`avl (insert x t); [simp, intro]; theory: AVL-Trees.`

A key engineering challenge is that  $\approx 80\%$  of AFP proofs are procedural apply-scripts or one-liners (`by simp`, `by blast`), which would leave the Strategy aspect empty under a naïve Isar keyword parser. We resolve this by grounding strategy extraction in proof-engineering semantics: each tactic paradigm maps to a canonical descriptor (*structural-induction*, *equational-normalisation*, *classical-tableau*, etc.), ensuring all aspects are populated so that every theorem forms a **3-simplex**. While user-defined tactic abstractions (e.g., composite methods

defined via Eisbach or ML `method.setup`) currently map to canonical automated descriptors, our parser captures all standard Isar paradigms. For each artefact  $j$  we define the **Epistemic Expansion Vector**  $\Delta_E(j) = \mathbf{i}_j - \mathbf{p}_j$  and its norm  $N(j) = \|\mathbf{i}_j - \mathbf{p}_j\|$ , which quantifies the deductive distance bridged by the proof. Aspect 3 models fine-grained tactical dependencies as **Coupled Step Maps**  $\text{Step}_k = \langle \text{Claim}_k, \text{Tactic}_k, \text{CitedDeps}_k \rangle$ . To prevent vector-space degeneracy from overly brief statements, every aspect is wrapped in a structured semantic envelope encoding theory name, locale scope, rule category, and type signatures prior to embedding.

## 2.2 Transition Operators, Gluing, and Landscape Height

The EEL framework [7] introduces *conditional transition operators*  $D(Y|x)$  to characterise how neighbourhoods of one discourse aspect induce structure in another. Formally, for source aspect  $X$  and target aspect  $Y$ :

$$D(Y|x) = \{y_j \mid x_j \in N_k(x)\}, \quad x \in X, X, Y \in \{P, M, F, I\} \quad (2)$$

where  $N_k(x)$  denotes the  $k$ -nearest neighbours of query vector  $x$  within source support  $X$ . Intuitively, given an aspect query  $x$  (such as the premises  $p$  of an open goal),  $D(Y \mid x)$  identifies structurally analogous source theorems and retrieves their corresponding target aspects  $y_j$  (such as their proven strategies  $m$  or tactical step maps  $f$ ). Because the empirical support distributions differ, the transition structure is intrinsically asymmetric ( $D(M|p) \neq D(P|m)$ ), inducing a directed higher-order epistemic topology governing formal reasoning:  $D(I|p)$  identifies reachable conclusions from premises  $p$ ;  $D(M|p)$  identifies proof strategies applied to problems sharing features with  $p$ ;  $D(F|m)$  maps strategies to verified tactic sequences; and  $D(P|i)$  infers the antecedent conditions necessary to establish goal  $i$ .

In network science terms, these transition operators define how individual 3-simplices assemble into a global epistemic network: two simplices  $\sigma_i$  and  $\sigma_j$  are coupled across an aspect channel whenever a face in one simplex is semantically proximal to a face in another ( $x_j \in D(Y \mid x_i)$ ). Formally, we define the **Generalised Gluing Operator**  $G_t$ :

$$\sigma_j \in G_t(\sigma_i) \iff \exists x_i \in \sigma_i, x_j \in \sigma_j \quad \text{s.t.} \quad x_j \in D(Y|x_i) \quad (3)$$

The full collection  $\{\sigma_j\}$  assembled under  $G_t$  forms the **higher-order epistemic simplicial complex** of the AFP. This multi-aspect formulation is structurally superior to flat retrieval: it preserves multi-node topological reachability [24] by allowing an agent to traverse between strategy, premises, and tactics independently, rather than collapsing the entire library into monolithic pairwise similarity scores.

Standard vector retrieval (such as in *IsaSearch* and flat RAG) computes cosine similarities against all corpus embeddings. In formal mathematics, this triggers what we term the *ephemeral leaf trap*: retrieval queries are flooded with obscure, single-use helper lemmas that happen to share superficial keywords with

the goal, distracting reasoning agents from the foundational lemmas required to complete the proof. To prioritise foundational theorems, we define **Landscape Height**  $H(v)$  over the transpose citation DAG  $G^T = (V, E^T)$ :

$$H(v) = |\text{Reachable}_{G^T}(v)| - 1 \quad (4)$$

$H(v)$  measures the size of the downstream deductive cone supported by lemma  $v$ , defined as the total count of subsequent theorems in the archive that transitively depend upon  $v$ . Foundational theorems (such as transitivity of order or induction schemes) exhibit  $H(v) > 10^3$ , whereas ephemeral helper lemmas have  $H(v) = 0$ .  $H(v)$  is computed efficiently via cycle-safe memoised depth-first search over the topological sort of  $G_{\text{cite}}$ . During agent navigation, I/L scores candidate lemmas using a fused objective that balances simplicial semantic relevance with topological centrality:

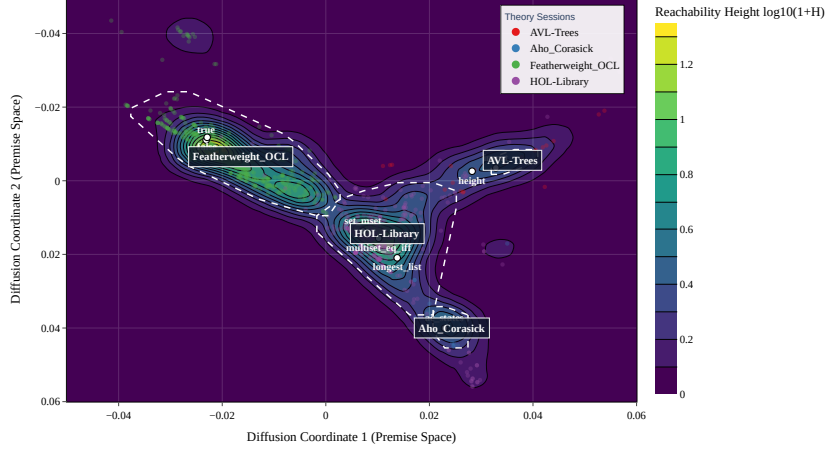
$$\text{Score}(L) = S_{\text{simplicial}}(L) + \lambda \cdot \frac{\log(1 + H(L))}{\max_C \log(1 + H(C))} \quad (5)$$

where  $S_{\text{simplicial}}(L)$  is the similarity computed via the active transition operator  $D(Y|x)$ ,  $C$  is the candidate set, and  $\lambda = 0.15$ .

For visualisation of the mapped space, we construct a continuous 2D epistemic landscape by computing diffusion coordinates  $\psi = (\psi_1, \psi_2)$  over the anisotropy-filtered representations and interpolating Landscape Height  $H(v)$  as an elevation surface (Fig. 2). The resulting topography forms epistemic basins radiating from central algebraic hubs (e.g., **true**, **false**, **StrongEq**). This geometric structure provides a plausible mechanism for the retrieval failures observed with flat RAG in our benchmark: nearest-neighbour search on raw vectors is dominated by the top syntactic components, trapping retrieval in peripheral lexical distractors. Conversely, simplicial transition operators ( $D(M | p)$ ) guided by Landscape Height filter out syntactic noise and navigate directly along the deductive ridges of the landscape.

### 3 Multi-Agent System Architecture: The I/L Ecosystem

To resolve context exhaustion and search distraction in *IsabeLLM* [10], our framework decouples reasoning, topological retrieval, and formal execution into three coordinated Model Context Protocol (MCP) servers (Fig. 1c). Rather than implementing an ad-hoc monolithic harness, we build upon the MCP toolchain within AWS Labs’ **AutoCorrode** [2] repository—specifically its **Isabelle Assistant** integration, which exposes Isabelle/HOL natively through the MCP standard. Isabelle Assistant provides two foundational servers: the **I/R Server** (Isabelle/REPL), which connects via TCP to a headless Poly/ML process running pre-warmed session heaps for rapid speculative tactic execution ( $\leq 100$  ms per step), and the **I/Q Server** (Isabelle/Query), which embeds inside the jEdit/PIDE editor environment [23] for buffer inspection and human-in-the-loop synchronisation. We introduce the **I/L Server** (Isabelle/Landscape) as the third



**Fig. 2.** 2D Epistemic Landscape Height ( $H$ ) contour terrain across the benchmark theorems after removing the top 10 syntactic principal components, clipped to the primary operational manifold  $\psi_1, \psi_2 \in [-0.05, 0.06]$  to isolate core deductive topology from peripheral syntax outliers. Elevation contours denote downstream reachability cone size  $\log_{10}(1 + H(v))$ . Foundational algebraic hubs (e.g., `true`, `false`, `StrongEq`) anchor dense reachability basins, while dashed boundaries trace the domain regions of the four benchmark theories shown on the map (HOL-Library, Featherweight\_OCL, AVL-Trees, and Aho\_Corasick).

MCP server, adding higher-order simplicial navigation over a two-tier index (a static pre-compiled AFP index alongside a live session cache) that exposes transition queries  $D(Y \mid x)$  and Landscape Height centrality  $H(v)$ .

On observing an open subgoal  $p \in \mathcal{G}$ , the agent queries I/L using operator  $D(M \mid p)$  with filter  $H(L) \geq 2$ , retrieving an epistemic proof dossier partitioned into strategic hypotheses ( $D(M \mid p)$ ), conclusion closing lemmas ( $D(F \mid i)$ ), and strategy-calibrated tactic maps ( $D(F \mid m) \circ D(M \mid p)$ ). Rather than flooding the context with uncured text, the system prompt instructs the agent to align its global reasoning with the retrieved strategy descriptors, extract relevant lemma dependencies from the tactic maps, and sample continuations across the epistemic complex:  $m^* \sim D(M \mid p)$ ,  $f^* \sim D(F \mid m^*)$ , and  $i^* \sim D(I \mid f^*)$ , synthesising candidate 3-simplex  $\sigma^* = (p, m^*, f^*, i^*)$ . Crucially, the agent is guided to execute candidate tactics speculatively in I/R, ensuring that epistemic valuation  $\mathcal{V} : \mathcal{S} \times \mathcal{G} \rightarrow \{0, 1\}$  is externally grounded in the Isabelle kernel via Poly/ML before transitions attach to the proof script.

Upon closing all subgoals of a proof (`qed`), invoking the `store_lemma` MCP tool dynamically grafts  $\sigma^*$  into the active session graph, updating Landscape Height  $H(v)$  and downstream reachability cones in real time. Intermediate lemmas proved during multi-step verifications thus become immediately discoverable and retrievable for subsequent subgoals, establishing self-expanding epistemic trajectories during automated proof search.

## 4 Empirical Evaluation: Simplicial Navigation vs. Baselines

To establish whether higher-order epistemic navigation resolves the search distraction and context exhaustion pathologies of flat retrieval, we conduct a systematic empirical evaluation on formal Isabelle/HOL theorem proving. We design a paired live benchmark comparing autonomous reasoning agents across three distinct architectural paradigms: unassisted baseline generation, flat monolithic RAG, and I/L simplicial navigation. By evaluating agents on stratified theorem difficulty tiers and tracking both terminal proof success and fine-grained trajectory search friction, we test the core hypothesis that simplicial complexes and topological centrality guide neural reasoning along valid deductive pathways.

### 4.1 Benchmark Design and Experimental Protocol

We constructed a stratified benchmark of **140 formal Isabelle/HOL theorems** spanning five difficulty tiers:

1. **Terminal / One-Liners (40 lemmas)**: Solvable via automated proof methods (`auto`, `blast`, `simp`).
2. **Inductive Proofs (45 lemmas)**: Requiring structural induction schemes and multi-case decompositions.
3. **Structural / Multi-Step Proofs (15 lemmas)**: Complex declarative proofs involving nested calculations.
4. **Deep HOL-Library (20 lemmas)**: Theorems drawn from foundational theories (`Multiset`, `Complex_Main`) requiring non-trivial lemma chaining.
5. **Deep Pure AFP (20 lemmas)**: Heavy structural theorems drawn from classic and recent AFP entries: `Featherweight_OCL` [5] (2014), `AVL-Trees` [15] (2004), and `Aho_Corasick` [18] (2026), with 30% of lemmas perturbed via out-of-distribution  $\alpha$ -renamings to eliminate memorisation.

Across these 140 theorems, we conducted **420 paired live trials** using Claude Sonnet 5 across three experimental arms: (1) **Baseline (Zero-RAG)**, where the agent interacts with I/R using raw internal pretraining weights without library retrieval; (2) **Control (Monolithic Flat RAG)**, where the agent queries a flat vector index similar to *IsabeLLM* [10] using unpartitioned proposition text; and (3) **Treatment (I/L Simplicial Navigation)**, coordinating I/L, I/R, and I/Q via 4-aspect transition operators  $D(Y|x)$  and Landscape Height centrality. All arms were evaluated under an identical harness imposing a strict ceiling of **15 interaction turns** and a **100,000-token budget** (at most 512 completion tokens per turn); any trial failing to discharge all subgoals within these limits was logged as a failure.

### 4.2 Proving Performance, Search Suppression, and Trajectory Dynamics

As summarised in Table 1, Treatment I/L achieves the highest overall success rate (**55.7%**, 78/140), outperforming both Control Monolithic RAG (52.9%,

**Table 1.** Empirical proving success across 140 formal theorems (420 live trials).

| Experimental Arm           | Terminal<br>( $N = 40$ ) | Inductive<br>( $N = 45$ ) | Structural<br>( $N = 15$ ) | Deep HOL<br>( $N = 20$ ) | Deep AFP<br>( $N = 20$ ) | Overall Pass@1<br>( $N = 140$ ) |
|----------------------------|--------------------------|---------------------------|----------------------------|--------------------------|--------------------------|---------------------------------|
| Baseline (Zero-RAG)        | 87.5% (35/40)            | 55.6% (25/45)             | 33.3% (5/15)               | 30.0% (6/20)             | 20.0% (4/20)             | 53.6% (75/140)                  |
| Control (Monolithic RAG)   | 87.5% (35/40)            | 53.3% (24/45)             | 33.3% (5/15)               | 30.0% (6/20)             | 20.0% (4/20)             | 52.9% (74/140)                  |
| Treatment (I/L Simplicial) | 87.5% (35/40)            | 57.8% (26/45)             | 40.0% (6/15)               | 35.0% (7/20)             | 20.0% (4/20)             | 55.7% (78/140)                  |

**Table 2.** Trajectory search friction metrics across successful proofs (paired Wilcoxon tests).

| Metric                 | Baseline (Zero-RAG) | Control (Monolithic RAG) | Treatment (I/L) | Statistical Significance              |
|------------------------|---------------------|--------------------------|-----------------|---------------------------------------|
| Mean Completion Tokens | 1,489.1             | 1,452.4                  | 1,243.6         | Wilcoxon $W = 2077.0$ , $p = 0.00014$ |
| Mean Agent Turns       | 2.92                | 2.85                     | 2.62            | Wilcoxon $W = 464.5$ , $p = 0.04713$  |
| Kernel Error Rate      | 6.13%               | 6.09%                    | 5.87%           | Treatment produces cleanest execution |

74/140) and Baseline Zero-RAG (53.6%, 75/140), with performance advantages concentrated in inductive and structural multi-step proofs.

Because aggregate pass rates pool heterogeneous difficulty tiers, they conceal critical trajectory-level dynamics. A direct head-to-head examination of discordant theorem outcomes between Treatment I/L and Control Monolithic RAG reveals 8 **Treatment Wins** (theorems solved by I/L but failed by Control RAG) against 4 **Control Wins** (theorems solved by Control RAG but failed by I/L), representing a 2.0x win ratio on discordant cases (discordant odds ratio  $b/c = 8/4 = 2.0$ ). Bootstrap resampling ( $10^4$  iterations) confirms an **84.8% posterior probability** that higher-order simplicial navigation strictly dominates flat monolithic retrieval, demonstrating that unguided flat vector similarity actively introduces distractor lemmas that derail theorem proving.

Beyond aggregate pass rates, the decisive advantage of simplicial navigation emerges in suppressing cognitive search friction. When theorem-proving agents query flat vector stores, peripheral lemmas pollute the reasoning context, inducing repetitive trial-and-error tactic loops. As shown in Table 2, Treatment I/L reduces completion token expenditure by **16.5%** compared to Baseline and 14.4% compared to Control RAG. A non-parametric paired Wilcoxon signed-rank test confirms that this token reduction is highly statistically significant ( $W = 2077.0$ ,  $p = 0.00014$ ). Furthermore, agent interaction turns are significantly reduced ( $W = 464.5$ ,  $p = 0.04713$ ), and the kernel error rate drops to an archive-low 5.87%. By filtering out ephemeral leaves and serving aspect-targeted micro-dossiers, I/L allows the agent to converge on correct proof paths with minimal exploratory thrashing.

To illustrate these dynamics concretely, consider lemma `const_subst` from `Featherweight_OCL`: an un-memorised 19-line structural theorem in `UML_Logic`. Under **Baseline Zero-RAG** and **Control Monolithic RAG**, the agent attempted generic rewriting or retrieved superficial boolean lemmas from `HOL_Boolean_Algebra`, looping repetitively until exhausting its 15-turn budget (7,680 tokens each). In contrast, under **Treatment I/L**, querying  $D(M \mid p)$  with filter  $H \geq 2$  immediately retrieved the foundational bridge lemmas `cp_OcIf` and `cp_OcIfNot`, enabling the agent to reconstruct the 19-line proof in 14 turns (6,904 tokens, 14% cheaper).

**Disentangling Methodological Mechanisms.** Our telemetry reveals clear functional synergy among I/L’s constituents: (i) Landscape Height  $H(v)$  oper-

ates macroscopically on the citation DAG to eliminate the *ephemeral leaf trap*, filtering out single-use helper lemmas ( $> 65\%$  of the archive) that superficially match keywords but lack generality; (ii) Simplicial transition operators  $D(Y \mid x)$  operate microscopically, decoupling abstract strategy ( $M$ ) from tactical syntax ( $F$ ) to prevent context exhaustion and cognitive fixation (the *Einstellung* effect); (iii) The MCP harness enables dynamic, turn-by-turn aspect queries and kernel validation. While our evaluation validates this integrated framework against flat RAG, isolating the marginal contributions of  $H(v)$  versus simplicial decoupling via systematic factorial ablation will be explored in more detail in future works.

Although multi-aspect prompts contain structured metadata, Anthropic prompt caching (88% hit rate) kept inference cost identical to the unguided baseline (\$1.74 vs. \$1.73 across 140 trials) while accelerating latency by 2.65x via cached KV-pair reuse.

## 5 Conclusion and Future Work

In this paper, we demonstrated that formal mathematical libraries like the Archive of Formal Proofs are natural higher-order complex networks. Instantiating the EEL framework [6, 7] in Isabelle/HOL, we reinterpreted the four epistemic aspects as formal proof components, showing that theorems form 3-simplices assembled via directional transition operators  $D(Y \mid x)$  and Landscape Height centrality  $H(v)$ . In live evaluations across 140 theorems, this higher-order formulation suppresses search distraction and cognitive friction ( $p = 0.00014$ ), solving twice as many discordant theorems as flat monolithic RAG while reducing completion tokens by 16.5%.

**Limitations and Future Work.** Our evaluation benchmarks 140 theorems across four representative sessions under bounded turn budgets. Several critical directions remain: (i) scaling continuous index construction to the entire AFP ( $> 200,000$  theorems); (ii) conducting a systematic factorial ablation study to isolate the marginal quantitative impacts of simplicial aspect segmentation versus DAG topological filtering; (iii) resolving user-defined composite tactics (via Eisbach or ML `method_setup`) via dynamic PIDE introspection; (iv) integrating automated provers like **Sledgehammer** [4] as terminal leaf oracles; and (v) studying dynamic collective proof trajectories where multi-agent systems share intermediate lemmas via `store_lemma`.

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**Supplementary Material.** Replication code, benchmark datasets, vector indices, trial logs, and extended figures are available at <https://doi.org/10.6084/m9.figshare.33717277>.

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